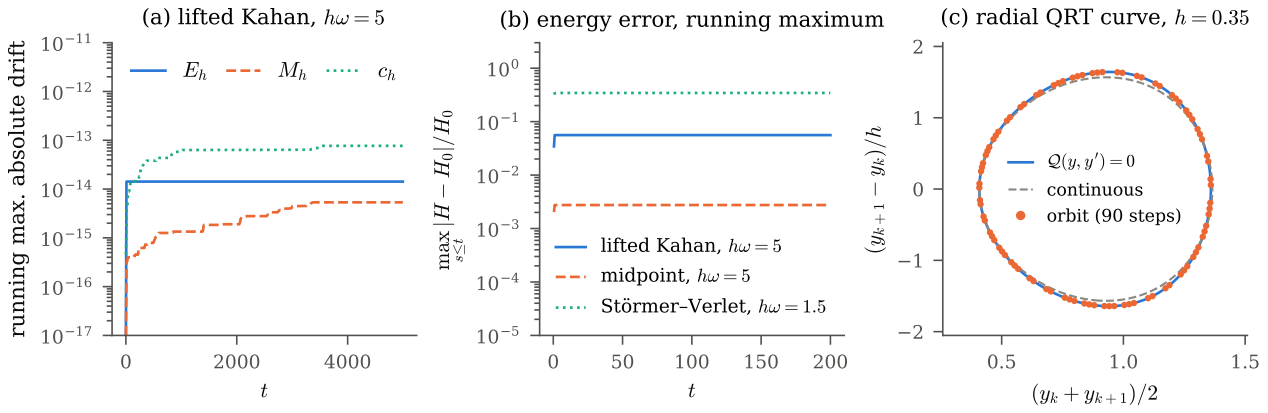

(a) lifted Kahan, $h\omega = 5$
running max. absolute drift
E_h
M_h
c_h
10^{-11}
10^{-12}
10^{-13}
10^{-14}
10^{-15}
10^{-16}
10^{-17}
0
2000
4000
t
(b) energy error, running maximum
$\max_{s \le t} |H - H_0|/H_0$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
lifted Kahan, $h\omega = 5$
midpoint, $h\omega = 5$
Störmer–Verlet, $h\omega = 1.5$
0
50
100
150
200
t
(c) radial QRT curve, $h = 0.35$
$(y_{k+1} - y_k)/h$
2
1
0
−1
−2
$\mathcal{Q}(y, y') = 0$
continuous
orbit (90 steps)
0.5
1.0
1.5
$(y_k + y_{k+1})/2$